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On-street parking for freight, services, and e-commerce traffic in US cities: A simulation model incorporating demand and duration

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ABSTRACT

This paper develops a modeling framework to estimate the on-street parking demand for urban commercial traffic in the US using publicly available data. The commercial traffic includes the freight trip generation (pick-ups and deliveries), service trip attraction (service visits to establishments), and e-commerce deliveries, altogether referred to as freight and service activities (FSA) in this study. The methodology includes a simulation model incorporating FSA demand estimation, probability distributions for FSA duration, and industry sectors at the 2-digit North American Industry Classification System (NAICS) level. The authors validated the model through a case study of the SoHo commercial complex's parking demand in New York City (NYC) (Campbell et al., 2018). In addition, the study analyzes the magnitude of the parking issue in NYC comparing five ZIP codes in Midtown Manhattan, which generate the highest commercial demand in the city. The study extends the analysis to the busiest ZIP codes in six cities in the US (Austin, TX; Columbus, OH; San Jose, CA; Seattle, WA; Washington DC; Boston, MA). The results show that the parking spaces required during peak hours at the denser areas (i.e., ZIP codes in Manhattan) cannot be accommodated by the available on-street parking capacities. The authors tested four demand management initiatives to reduce the peak-hour needs. The off-hours strategy, which indicates 30 % of freight pick-ups and deliveries and 51 % of service visits shifting to the off-hours, is the most effective one, with the potential to reduce daytime parking needs by 30 %. Results also show the need to dedicate curb space for FSA vehicle parking to satisfy time-of-day requirements.

1. Introduction

Cities are significant generators of commercial activity and are the principal drivers of a country's economy. The daily commercial transactions also bring a considerable amount of traffic that adds to the congestion, pollution, safety issues, and other adverse outcomes that hamper the community's livability. Impeding this influx of commercial vehicle traffic is impossible, as the daily transport of goods and services to individuals and businesses feeds the economy. City agencies and transportation planners must understand the magnitude of the problem to implement suitable policies that will help improve the economic viability of companies and the well-being of residents.

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Previous research that estimates trip generation for freight and service activity (FSA) in urban areas indicated the significant impact of the daily economic activities in establishments and households (Holguín-Veras et al., 2018a; Holguín-Veras et al., 2018c). FSA includes the vehicle traffic generated by freight pick-ups and deliveries and the service visits to establishments. Freight trip generation (FTG) models estimate the daily freight vehicles produced (FTP) and attracted (FTA) by these activities. The service trip attraction (STA) models estimate the daily service vehicles generated due to the demand for services at the establishments.

Among these two forms (freight and service) of commercial activity that comprise FSA, the one that has received the most attention is the freight trips associated with deliveries and pick-ups. The burst of FTG publications in recent years is a result of this phenomena (Iding et al., 2002; Kawamura et al., 2008; Davydenko et al., 2011; Holguín-Veras et al., 2011a; Kawamura and Miodonski, 2011; Holguín-Veras et al., 2012; Lawson et al., 2012; Holguín-Veras et al., 2013; Jaller et al., 2013b; Günay et al., 2016; Alho and de Abreu e Silva, 2017; Gonzalez-Feliu and Peris-Pla, 2017; Holguín-Veras et al., 2017a; Gonzalez-Calderon et al., 2018; Gonzalez-Feliu et al., 2018; Gonzalez-Feliu and Sánchez-Díaz, 2018; Holguín-Veras et al., 2019; Pani et al., 2019; Puente-Mejia et al., 2019; Dhulipala and Patil, 2020; Holguín Veras et al., 2020; Pani et al., 2020). In contrast, studies focusing on the service traffic are not as predominant (Allen and Agnello, 2004; Ellison et al., 2017; Holguín-Veras et al., 2020; Caron and Holguín-Veras, 2022). These studies are not as common because the proportion of traffic related to service is minimal compared to freight. Also, due to the size of the service vehicles—which are mostly comprised of vans and small trucks—they tend to cause less impact than large trucks blocking the roads and intersections. Not considering service traffic within the urban traffic environment is a major problem however, because although they are less prevalent, they occupy significant curbside space because of their longer parking times. The authors' previous research on service trip attraction indicated that "service vehicles generate between 22.7 % and 78.4 % of the total parking requirements, because of the effects of the duration times, while they only represent 6.32 %–24.84 % of the total commercial traffic" (Holguín-Veras et al., 2020).

Given the accumulation of daily traffic due to FSA in city streets, it is imperative to provide adequate parking spaces for these commercial vehicles. City planners and transportation authorities should address the lack of parking spaces in urban centers, including the limited space available for service and freight vehicles at the curb. An analysis of commercial parking durations is necessary to address this correctly.

Recent research on commercial vehicle parking has shown interest in the integration of demand and parking duration for FSA and its relevance to urban transportation management (Allen et al., 2008; Browne et al., 2010; Dezi et al., 2010; Alves et al., 2017; Muñuzuri et al., 2017; Oliveira et al., 2017; Campbell et al., 2018; Schmid et al., 2018; Girón-Valderrama et al., 2019; Dalla Chiara and Goodchild, 2020). As duration data becomes available, simulation-based methodologies are valuable to predict a more precise accumulation of FSA vehicles that require parking. The latter can benefit city planners to make more informed decisions for a better allocation of parking spaces.

This paper studies the space requirements of FSA vehicles by time-of-day using parking duration distributions for both freight and service. The authors analyze parking duration data obtained for freight pick-ups and deliveries recently studied in New York City (Schmid et al., 2018). This paper also uses the analysis of service durations obtained from the authors' previous research (Holguín-Veras et al., 2020) to study service vehicle parking.

The authors propose a simulation model that inputs the estimates of daily FSA vehicles by industry sector, the time-of-day patterns for both freight and service vehicles, and the duration of service parameters. The authors use the North American Industry Classification System (NAICS) to classify industry sectors. The authors' previous research (Holguín-Veras et al., 2017a) classifies commercial establishments into Freight and Service-Intensive sectors. Freight-Intensive Sectors (FIS) are the industry sectors for which the production and consumption of freight are essential to their economic activity. While the Service-Intensive Sectors (SIS) are those for which the provision of services is their principal activity. The list of sectors in FIS and SIS based on 2-digit NAICS (in parenthesis) is as follows:

- Freight-Intensive Sectors (FIS): Agriculture, Forestry, Fishing and Hunting (11); Mining, Quarrying, and Oil and Gas Extraction (21); Utilities (22); Construction (23); Manufacturing (31–33); Wholesale Trade (42); Retail Trade (44–45); Transportation and Warehousing (48–49); Accommodation and Food Services (72).
- Service-Intensive Sectors (SIS): Information (51); Finance and Insurance (52); Real Estate and Rental, and Leasing (53); Professional, Scientific, and Technical Services (54); Management of Companies (55); Administrative, Support, Waste Management (56); Educational Services (61); Health Care and Social Assistance (62); Arts, Entertainment, and Recreation (71); Other Services (81); and Public Administration (92).

The model simulates the parking durations of each FSA vehicle, which includes the duration of service and walking times to and from the parking locations. FSA duration follows a log-normal distribution with parameters estimated from the empirical data for freight pick-ups and deliveries (Schmid et al., 2018), and service visits to establishments (Holguín-Veras et al., 2020). The simulation model uses uniformly distributed walking times, which range between 1 and 2.5 min (one-way trip) for freight pick-ups and deliveries, as well as for service visits.

The authors test the simulation model through a small case study to observe the FSA parking requirements at the street level. The authors use data collected from commercial establishments from the SoHo Business Improvement District in Lower Manhattan, New York. This paper expands the research from Holguín-Veras et al. (2017a) by developing a simulation model that incorporates FSA durations and estimates the number of trips divided by time-of-day.

The parking simulation model developed in this research is also implemented on larger networks. The study compares the top five ZIP codes that generate the most commercial traffic in NYC. Similarly, the study compares six cities in the US with a population larger

than 100, 000 (Austin, TX; Columbus, OH; San Jose, CA; Seattle, WA; Washington DC; Boston, MA). The authors also selected the most congested ZIP code (i.e., with the highest FSA) from each city. For each city ZIP code, the FSA vehicle trips were computed by applying the FTG/STA models developed by Holguín-Veras et al. (2017a) on the establishment data at the ZIP code-level from the County Business Patterns database (U.S. Census Bureau, 2018a). In addition to freight and service trips to commercial establishments, the FSA estimates also include the Business-to-Consumer (B2C) deliveries, based on the authors' estimates with data on online purchases from the National Household Travel Survey (McGuckin and Fucci, 2018).

This research quantifies the FSA parking needs at the ZIP code per time-of-day. The authors compare the FSA parking road length requirements estimated by the simulation model with the on-street parking capacities at these ZIP codes. Thus, the simulation model estimates the FSA parking demand (i.e., number of vehicles, curbside road space) by the time-of-day and the occupancy rates (proportion of total curb space occupied by FSA parking).

This paper compares the parking demand for four scenarios of FSA by time-of-day. The first scenario is "as is" (base case) which is nothing but the current situation. The other three scenarios consider demand management strategies that can mitigate the effects of peak-hour traffic congestion and reduce parking demand during specific times of the day. The second scenario evaluates shifting a proportion of FSA trips to the off-hours (7 PM–6 AM). The percent of freight pick-ups and deliveries shifted to the off-hours is obtained based on the empirical findings from carriers in New York City regarding their willingness to participate in the off-hours (Holguín-Veras et al., 2018b). For service providers, the percent shift is derived based on the authors' previous findings of the establishments' willingness to receive services during the off-hours (Holguín-Veras et al., 2021; Caron and Holguín-Veras, 2022). A third scenario considers staggered deliveries or services, i.e., an equal proportion of trips during the daytime. A fourth scenario considers the receiver-led consolidation strategies that reduce freight traffic between 20 and 30 % (Holguín-Veras and Sánchez-Díaz, 2016).

The rest of the paper is organized as follows. Section 2 summarizes the literature review related to curbside parking for commercial vehicles. Section 3 presents the analysis of the duration data of both service and freight vehicles. Section 4 presents the simulation model developed to estimate on-street parking needs by time-of-day. Section 5 discusses the results of implementing the model at the street and ZIP-code levels. Section 6 presents the policy implications derived from the analyses. Section 7 summarizes the key research findings produced in the paper.

2. Literature review

Studies related to on-street or curbside parking for commercial vehicles are becoming more relevant and popular today, given the issues and traffic associated with limited parking in urban environments. On-street parking involves *"all metered and unmetered parking along the roadsides"* (Asiyanbola and Akinpelu, 2012). Studies are necessary to understand the magnitude of the problem. In the case of commercial vehicles, the issue involves congested streets and detrimental conditions in the delivery of goods and services. The latter makes travel much more expensive due to longer dwelling times, the addition of parking fines, and other costs related to the delays.

The lack of capacity, particularly in the downtown areas of large cities, is not a recent issue. More than fifteen years ago, Arnott acknowledged the problem. He explained, *"In the downtown areas of almost all major cities, on-street parking is used to capacity throughout the day, and in many of them, the on-street parking capacity utilization rate exceeds 100 % due to double parking, illegal curbside parking, and parking on the sidewalk. On-street parking under these conditions is said to be saturated."* (Arnott, 2006).

Freight surveys developed over the years consider vehicle parking aspects, like dwell times and loading and unloading durations. Allen et al. (2008) reviewed the results from 30 urban freight studies conducted in the UK from 1996 to 2008. Browne et al. (2008) explain that 23 of these studies that collected dwell times from establishment and vehicle observation surveys estimated average dwell times—the time vehicles park while loading/unloading goods—between seven and 34 min. These studies also analyzed loading and unloading activities, where on-street stopping locations caused more traffic delays than off-street. Yet, off-street areas are not always available, and further delays are caused by difficulty maneuvering vehicles in and out of the parking facilities. On-street deliveries occurred in more than 50 % of the cases, as reported in 12 of the 17 surveys.

Research in the last decade evidenced numerous publications related to on-street commercial parking in different cities worldwide. A study in Bologna, Italy, estimated that freight vehicles' loading/unloading process takes an average of 14 min (Dezi et al., 2010). Jaller et al. (2013a) studied freight vehicle on-street parking in some of the most congested ZIP codes in New York City. The latter study estimated daily vehicle trips using FTG models and on-street parking during peak hours. Their results indicate insufficient road capacity at the busiest ZIP codes in the city. Beziat (2015) studied the spatial and temporal distribution of delivery vehicles in Paris. The author contrasted these estimates with the actual supply of parking spaces. The results show that the current capacities are barely sufficient during peak hours to accommodate the parking of freight vehicles. Muñuzuri et al. (2017) studied the loading zones' availability for retail establishments for streets networks in Seville, Spain. They used data collected from a retailer survey that considered aspects of the delivery, such as the frequency and time-of-day of the delivery. Oliveira et al. (2017) collected FTG data of the pubs and restaurants sector in Belo Horizonte, which included the time of delivery and duration of unloading activities. Alves et al. (2017) collected FTG data from commercial establishments in the historic center of São João Del Rei in Brazil using FTG models. Silva et al. (2020) estimated the demand for loading and unloading freight vehicles in this Brazilian city's historic center. The authors estimated an average of 710 daily FTG, which requires at least 43 spaces for loading and unloading. As the center has only eight available spaces, representing 18 % of total demand, this study proposes new locations. The paper suggests transportation demand management measures that could be used in conjunction with the allocation of new parking spots.

Campbell et al. (2018) studied the freight and service parking needs using the FTG/STA models in Holguín-Veras et al. (2017a). The authors analyzed two case studies: a small commercial center in Troy, NY, and two blocks of a busy commercial district in Lower Manhattan in NYC. The study considered minimum, average and maximum durations to analyze parking needs during the peak-hour

periods. Escand et al. (2018) studied the commercial-dedicated parking space to accommodate home deliveries to residential and high-density neighborhoods in NYC. Schmid et al. (2018) collected parking duration data of 183 freight vehicles in several locations in New York City. Nearly 80 % of the vehicles were parked in illegal locations. Of those illegally parked, 60 % were double-parked. Half of the vehicles had no possible legal parking location within one block. Girón-Valderrama et al. (2019) collected parking data at one of the densest areas in downtown Seattle. They observed commercial vehicles parked at the curb and the loading/unloading zones. The study captured vehicle types' classification, dwell times, and parking choice of curb use in authorized and unauthorized spaces. Dalla Chiara and Goodchild (2020) studied GPS data of 2900 commercial vehicle trips to estimate the time commercial vehicles spent searching for a parking space (cruising) in downtown Seattle. They estimated an average estimated parking cruising time of 2.3 min per vehicle trip.

Past research also shows simulation models developed to capture the behavior of commercial vehicle parking in city streets. Munuzuri et al. (2002) developed a dynamic freight vehicle parking model using a microscopic traffic simulator, TRAMOS. The arrival of both private and freight vehicles is modeled through an O-D generation model. The authors considered the types of parking facilities, including legal (i.e., curbside, loading/unloading zones) and illegal parking areas (i.e., double parking, parking on the sidewalk). The model also considers parking choice and the number of stops per vehicle type. Nourinejad et al. (2014) developed a traffic simulation that considers a parking choice model that selects suitable parking facilities and locations. The results show the impact of on-street parking in a congested street system at Toronto's central business district (CBD). Nourinejad and Roorda (2017) developed a model that studies the behavior of rational carriers concerning parking policy under steady-state equilibrium conditions. The authors introduce in their model illegal parking behavior elasticity for dwell times, enforcement levels, parking citation fines, and citation probability. Dalla Chiara et al. (2020) developed a discrete-choice model for commercial vehicle parking in urban areas using data collected from two urban areas in Singapore.

The existing literature on commercial parking demand is heavily based on the estimation of freight demand or the simulation models combined with the data from the parking devices. Also, no research combines the estimation of commercial demand (FSA) and parking, as the latter requires durations of both freight and service vehicles. There are very few studies that account for service vehicle traffic. Without acknowledging the parking of service vehicles, the urban parking and commercial traffic picture is incomplete. The most recent estimates are presented by the authors' previous research (Holguín-Veras et al., 2017a; Holguín-Veras et al., 2020). Regarding the literature's simulation models that estimate commercial parking demand, none of them combine the establishment-level trip estimates and the duration analysis that facilitate FSA analysis with publicly available data. The modeling approach developed in this research is unique. It paves a way to study the commercial parking demand for the entire US at the ZIP-code level using publicly available data (U.S. Census Bureau, 2018a).

The contribution of this paper is threefold. First, the authors integrate the establishment-level estimates of daily freight and service traffic, including the B2C, to estimate on-street parking using FTG/STA models from Holguín-Veras et al. (2017a). Second, parking durations are included for freight and services separately in the simulation model in the form of probability distribution functions that can better capture the reality of commercial parking demand. Finally, the authors use the model to evaluate several demand management policies that affect commercial vehicle traffic demand and parking space utilization. Results show that the model is applicable to larger networks. Its application at the ZIP-code level in dense urban areas in NYC and other urban centers in the US shows how the different initiatives help reduce parking needs during peak hours and improve on-street parking utilization.

3. Data analysis

To study the effects of FSA in commercial vehicle parking, the authors analyzed the duration data of both service and freight activities. The analysis below considered two datasets, one for service duration and the other for the parking duration of freight deliveries and pick-ups.

The service duration data were collected from commercial establishments in New York City (NYC) and the New York State Capital Region (CR). CR is the metropolitan area of Albany, the capital city of New York State. The surveys were collected in 2015 and included the number of daily freight deliveries and pick-ups, service trip visits per week, types of vehicles used, and the duration of the services, among other characteristics. The daily freight deliveries and pick-ups were used to estimate FTG, while the service trip visits at the establishments were used to estimate STA.

Freight parking durations were analyzed from Schmid et al. (2018)'s study of freight vehicle parking in NYC. Their data is part of the dataset from the City College of New York (CCNY) team as part of their commercial vehicle study in urban areas (Conway et al., 2016). The study considered 183 parking events directly observed in four neighborhoods of NYC (in Manhattan and the Bronx) during the period of April-May of 2015. Schmid et al. (2018) shared the freight parking data for the analysis produced in this section.

3.1. Duration of service activity

A service is a *"a business that offers a particular type of help or work"* (Cambridge English Dictionary, 2020). A service visit to commercial establishments refers to any maintenance or repair activity, such as air conditioning services, laundry, electricity, cable, etc. The duration of the service activity is the amount of time that the business took to perform the service at the establishment. The pattern of service durations is critical, as it is a major *"determinant of the throughput of the parking system"* (Holguín-Veras et al., 2020). This section considers the study of service duration patterns using the data collected in 2015 from 219 establishments and summarized by the authors in Holguín-Veras et al. (2020).

Table 1 shows the summary statistics of the service duration data for all establishments and at each metropolitan area (NYC and CR) for the freight-intensive (FIS) and service-intensive sectors (SIS). The statistics show that the duration of service activity is significantly

lower (37 %) for establishments in CR compared to NYC. For all establishments in NYC, service durations range from one to 600 min, with an average of 93.22 min. Additionally, the duration of service activities between FIS and SIS shows different behavior between NYC and CR. In NYC, the FIS (101.74) has about 18 % longer average service durations than SIS (85.84). At the same time, the SIS in CR has about 23 % longer durations than FIS.

Table 1 shows a high variability of the data reported for service duration with a coefficient of variation (CV) between 0.96 for FIS and 2.03 for SIS in CR. The standard deviation is about 108 min for both FIS and SIS in NYC. Variability is higher for the SIS establishments in NYC and CR, where CR shows higher differences in CV. The CV for the FIS establishments is almost equal to one for NYC and CR.

The median values provide additional insights into the service duration distribution patterns. For all establishment groups, the median is lower than the average value. The latter suggests that the data may not be normally distributed. Traditionally, the distributions used to model positively skewed data are the gamma and log-normal distributions (Marchenko and Genton, 2010). The authors tested both distributions, and the Log-Likelihood indicator is higher for the log-normal (−366.28) than gamma fTable (−359.76) distribution. Fig. 1 shows the normal probability plots that compare the data with the theoretical standardized normal distribution (Chambers et al., 1983) for the natural logarithmic (log) of service durations generated through STATA version 14.2 (Statacorp, 2016). Any deviation of the data points from the straight line in the figures shows the data that do not fit the normal distribution. The plots indicate that the log values of service durations are nearly normally distributed for all datasets.

3.2. Duration of freight pick-ups and deliveries

The duration of freight pick-ups and deliveries are analyzed from data collected by Schmid et al. (2018). The statistics correspond to the duration times reported for the 177 valid observations. Table 2 shows the summary statistics of the duration data analyzed.

The duration data of freight pick-ups and deliveries were collected only for the establishments in NYC, which is different from STA duration data, which consider both NYC and CR (the metropolitan area of New York State's capital city, Albany). Unlike the STA durations, Schmid et al. (2018)'s data are collected from the observations made on the parking of freight vehicles. These estimates include the time the freight carrier walks to and from the establishment and loading/unloading activities.

Similar to the STA durations, the data distribution for freight vehicles is positively skewed, with the median being much lower than the average duration of 16 min. Moreover, Schmid et al. (2018) observed 37 percent of the observations recorded five minutes or less, and 67 percent were less than 12 min. A few observations show durations between one and two hours long.

Fig. 2 shows the distribution plots for service (STA) and freight (FTG) duration data. Even though both datasets are not comparable, these plots indicate the longer durations evidenced by service visits. Notice that the service duration data (striped bars) do not include the time spent walking from the parking space to the establishment, and show much longer average durations and a more elongated distribution.

Although the distributions seem different, both observations are likely to be log-normally distributed. Additional plots explain this more accurately—the probability density plots (STATA, 2020b) in Fig. 3(a, b) show right-skewed duration patterns. Moreover, the normality plots (STATA, 2020a) of the log-values of durations suggest these are normally distributed. Additionally, the Shapiro-Wilk tests for freight and service duration data indicate that the log-transformed durations are normally distributed (Shapiro and Wilk, 1965). The latter suggests a log-normal distribution is a better fit for the duration patterns of FSA activity in the simulation model explained in the next section.

The mean and standard deviation statistics (Tables 1 and 2) obtained from the data above are transformed to the log-normal distribution parameters. However, note that the data on freight parking durations (in Table 2) include the pick-up or delivery at the establishment, loading/unloading, and the walking times between the vehicle and establishment. For comparison purposes with the data collected for service visits, the average values of freight parking duration data reported by Schmid et al. (2018) were adjusted to account for the walking times. For the analysis in this paper, the authors subtracted 2.5 min of average walking times. This value corresponds to the average of the uniformly distributed values for walking times between 1 and 5 min. The authors estimated this range based on how far a carrier is willing to park away from the establishment. Walking a distance of a long block (approximately 185 m) to deliver a package takes about 2.5 min for the delivery person. The authors assumed a walking speed of 4.5 km/h for freight pick-ups and deliveries (Muñuzuri et al., 2012). Table 3 shows the statistics used for the duration for both service visits and freight pick-ups and deliveries.

Table 1
Summary Statistics of Service Duration.

Statistics	NYC			CR			Total		
	FIS	SIS	All Sectors	FIS	SIS	All Sectors	FIS	SIS	All Sectors
Mean (min)	101.74	85.84	93.22	52.03	64.17	58.48	83.71	78.06	80.69
Std Deviation (min)	108.18	108.94	108.49	49.93	130.43	100.65	94.24	117.01	106.81
Coeff. of Variation	1.06	1.27	1.16	0.96	2.03	1.72	1.13	1.50	1.32
Median (min)	60.00	60.00	60.00	30.00	20.00	30.00	60.00	45.00	60.00
Min (min)	2	1	1	1	1	1	1	1	1
Max (min)	480	600	600	180	610	610	480	610	610
Sample Size	65	75	140	37	42	79	102	117	219

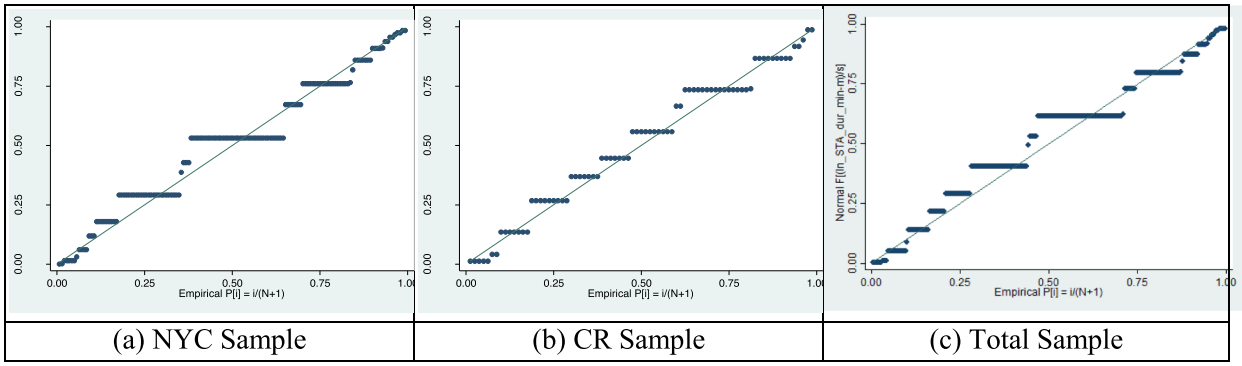


Fig. 1. Normality Plots of Service Duration Data.

Table 2
Summary Statistics of the Duration of Freight Pick-Ups and Deliveries.

Statistics	Mean (min)	Std Dev (min)	Median (min)	Min (min)	Max (min)	Sample Size
Values	15.66	21.50	7.00	1	125	177

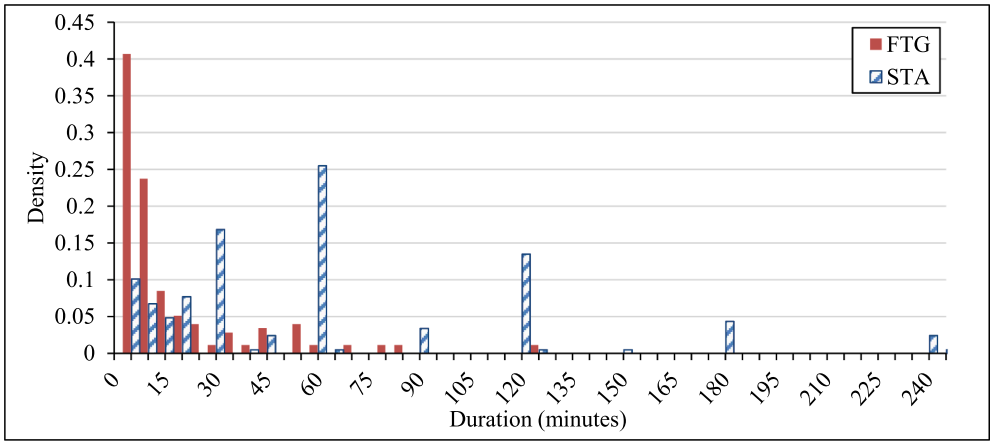


Fig. 2. Duration of Service Visits (STA) and Freight Pick-Ups and Deliveries (FTG).

4. Simulation model

The availability of the models to estimate FSA demand and their duration patterns provide a unique opportunity to develop quantitative techniques to estimate the parking needs of commercial vehicles at a given geographical level (e.g., ZIP code). In designing the methodology, the authors decided that it should be probabilistic in nature, considering the FSA estimated for the area under study and the duration patterns by industry sector (e.g., NAICS). A probabilistic technique is needed because it is the only way to account for the inherent variability in the FSA duration among industry sectors. That could be accomplished, for instance, through a simulation that computes the number of parking spaces required to accommodate service vehicles by time-of-day. The importance of considering the FSA generated at the study area is rather apparent, as curbside space allocation should be consistent with the local realities. The other factor is the service technicians' walking time between the parking location and their customers. The parametric modeling techniques for duration times such as Weibull, Lognormal regression models are not included as the purpose of this study is confined to utilizing publicly available data to estimate parking demand. Hence, the non-parametric probability distributions are a better way to include durations.

The approach suggested by the authors entails the computation of the parking duration that ensures availability of space for a preset percentage of visits, considering the FSA duration and walking time between the parking location and the customers, as shown in Equation (1). Note that the walking time to receiver, w_i, the time it takes from the parking slot to the receiver's establishment, is multiplied by two to account for both trajectories, to and from the establishment.

$$D_i = d_i + 2w_i \tag{1}$$

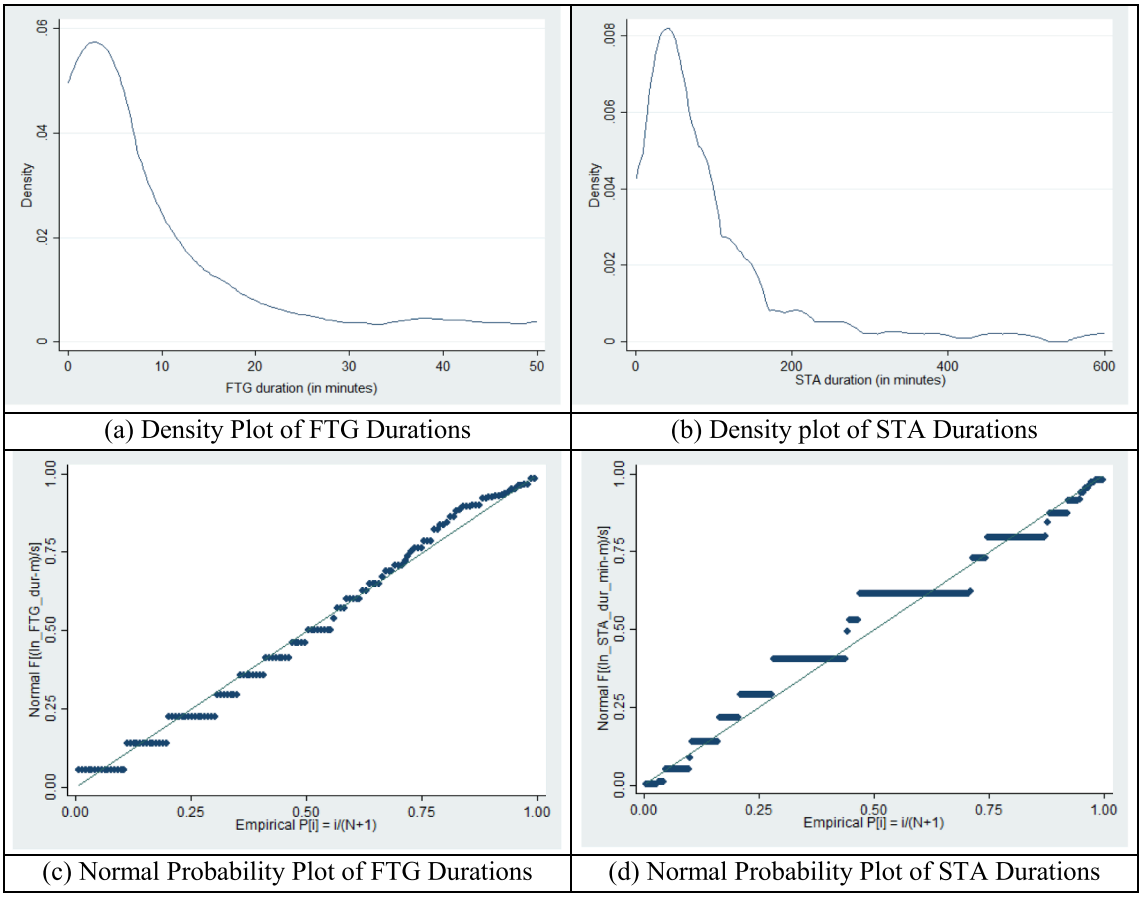


Fig. 3. Distribution Plots of Duration Data.

Table 3
Duration Statistics for Service Visits and Freight Pick-Ups and Deliveries.

Statistics	Service Visits									Freight Pick-Ups and Deliveries		
	NYC Sample			CR Sample			Total Sample					
	Mean	Std. Dev.	Obs	Mean	Std. Dev.	Obs	Mean	Std. Dev	Obs	Mean	Std. Dev	Obs
FIS	101.74	108.18	65	52.03	49.93	37	83.71	94.24	102	13.77	19.44	145
SIS	85.84	108.94	75	64.17	130.43	42	78.06	117.01	117	24.19	27.89	32
All Sectors	93.22	108.49	140	58.48	100.65	79	80.69	106.81	219	15.65	21.50	177

Where:
D_i: Parking duration by industry *i*.
d_i: Duration for FSA activity by industry *i*, $d_i L\tilde{o}gnormal(\mu_i^*, \sigma_i^*)$
μ_i^*: Log-normally transformed mean of FSA duration for industry sector *i*.
σ_i^*: Log-normally transformed standard deviation of FSA duration for industry sector *i*.
w_i: Time to receiver for industry sector *i*, $w_i \tilde{U}[w_{LB}, w_{UB}]$
w_{LB}: Lower bounds of walking times.
w_{UB}: Upper bounds of walking times.

4.1. Log-Normally transformed mean and standard deviations

The data analysis from Section 3 suggested the duration (d_i) of FSA trips are log-normally distributed. By definition, if d_i has a lognormal distribution, then $Z_i = \log(d_i)$ has a normal distribution. Given the μ_i be the mean and σ_i be standard deviation parameters of a normal distribution for $Z_i \tilde{N}(\mu_i, \sigma_i)$, the following equations (2)–(5) show the process of generating the mean and standard deviation of

respective log-normal distributed paramters for d_i. Using the transformed parameters, the generated random values follow the log-normal distribution.

Step 1. Calculate the variance v_i of the FSA durations:

$$v_i = \sigma_i^2 \tag{2}$$

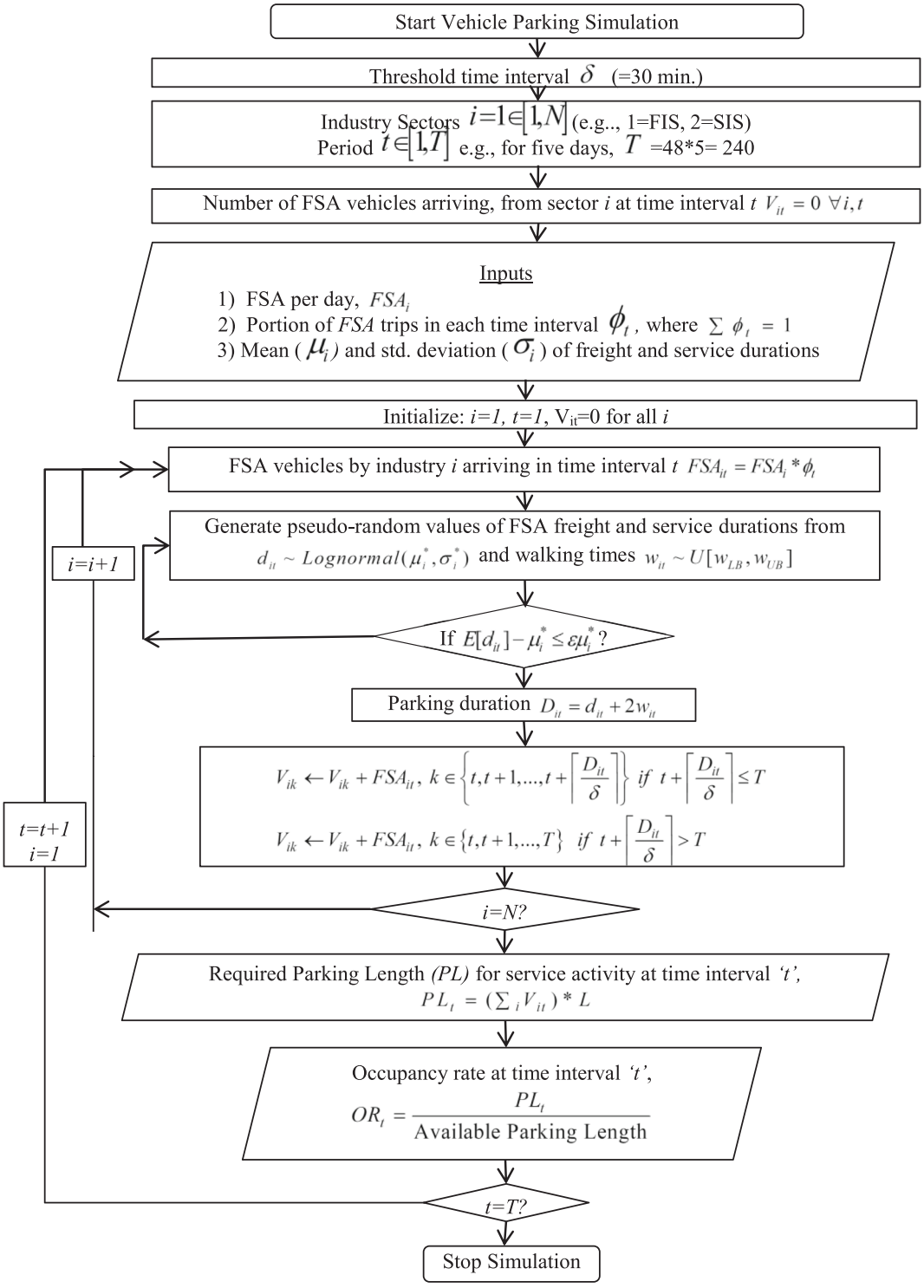


Fig. 4. FSA Vehicle Parking Simulation.

Step 2. Obtain the parameter θ_i to calculate the transformed values:

$$\theta_i = \sqrt{v_i + \mu_i^2} \tag{3}$$

Step 3. Obtain the transformed mean:

$$\mu_i^* = \log(\mu_i^2/\theta_i) \tag{4}$$

Step 4. Obtain the transformed mean:

$$\sigma_i^* = \sqrt{\log(\theta_i^2/\mu_i^2)} \tag{5}$$

Equations (4) and (5) are used to estimate the FSA durations, where $d_i \widetilde{Lognormal}(\mu_i^*, \sigma_i^*)$.

4.2. Walking times

For purposes of the simulation, and due to the lack of data about the distribution of walk times w_i, the authors decided to assume a uniform distribution with a specific interval for the walk time between the location of the parked vehicle and the establishment. This interval $[w_{LB}, w_{UB}]$ was selected based on the range of values suggested by freight carriers, as those best represent the realities of American cities. The lowest value represents the case where the carrier can drive in front of the service activity site to deliver a small package that can be easily retrieved, i.e., one minute. Thus, $w_{LB} = 1$. The upper value represents the maximum period of time that carriers or service providers are typically willing to allocate to make a single delivery. For freight vehicles, walking a distance of a long block (approximately 185 m) to deliver a package, for instance, may prevent them from completing the rest of the deliveries. Assuming the walking speed of 4.5 km/h for freight pick-ups and deliveries as suggested by Muñuzuri et al. (2012), a reasonable value for the upper bound is $w_{UB} = 2.5$ minutes. In the case of service providers, data is not available. Thus, for service visits, the authors assume the same parameter $w_{UB} = 2.5$ minutes.

4.3. On-street parking occupancy rates

One of the simulation model's outputs is on-street parking occupancy rates. At each time interval, the model computes the on-street parking occupancy rate by dividing the total road length requirements by the available parking length. Equation (6) shows the occupancy rate at time interval *'t'*,

$$OR_t = \frac{PL_t}{\text{Available Parking Length}} \tag{6}$$

Where PL_t is the total parking length obtained from the number of freight and service vehicles arriving at time t multiplied by the average vehicle length, L. The authors assume different average vehicle lengths for freight and service vehicles, as explained in the next section.

The Available Parking Length used to compute the parking on-street occupancy ratio is the "unchangeable" capacity of on-street road length in the area of study. The authors estimate these values from the street network with the help of GIS tools (TRANSCAD and Google Earth). First, the authors use the total road length obtained from the street network totals in TRANSCAD. Second, the available parking length is estimated by subtracting the total road length from the space for the intersections, crosswalks, fire hydrants, bus turnover lanes, and other dedicated lanes for turning movements.

4.4. Simulation procedure

Fig. 4 shows a schematic of the simulation procedure used to estimate the commercial vehicle parking requirements for FSA at a given period. The first step of this process is to define the threshold time interval δ, i.e., the minimum period for the analysis within which the parking demand for FSA is assumed to be constant. For instance, if the $\delta = 30$min (i.e., the day comprising a 24-hour period is divided into 48-time intervals, $T = 48$), the parking simulation estimates the parking requirements for FSA vehicles for each 30 min of the day by industry sector (NAICS, i), for a given geographic region. To model assumes that all vehicles for a given time period δ arrive at the start of the interval. Since the models used to estimate FSA in (Holguín-Veras et al., 2017a) provide trips per day, one must define a factor ϕ that converts the FSA per day to the corresponding FSA per time interval δ.

Another assumption is that the entire parking space is empty $V_{i,t=0} = 0$ at the beginning of the simulation, which is considered to be the first day at 12 AM. The simulation may run for several days to guarantee a steady-state analysis of the accumulation of vehicles over time. The authors simulated FSA parking demand for 30-minute intervals for five consecutive days i.e., T = 240 (48 × 5). While the analysis on FSA parking demand is carried out for the third day to avoid any under/over predictions.

The input data consists of: (1) daily FSA trips, i.e., the FTG and STA estimates per industry sector; (2) the factors ϕ_t that converts FSA per day to FSA per δ time period; and (3) freight and service duration parameters for the various sectors that result from the data analyzed in the previous section. The ϕ_t factors are the proportion of FSA that arrives at the given time interval t. Before running the simulation, the geographical area must be defined. At the geographical area, the estimates of FSA trips, traffic proportions, and freight and service duration statistics are determined.

The parking duration of FSA activity (D_{it}) per industry sector i, for a given time interval t is estimated using Equation (1). To obtain the parking durations, the simulation generates r pseudo-random numbers. Pseudo-random numbers of duration d_{it} follow the log-normal distribution or $d_{it}\tilde{L}ognormal(\mu_i^*, \sigma_i^*)$, and walking times to receiver (w_i) are uniformly distributed between w_{LB} and w_{UB} minutes or $w_{it}\tilde{U}[w_{LB}, w_{UB}]$, as shown in Equations (2)–(5). To guarantee that the values follow these distributions, random values are generated until the expected value of the last ten iterations is close to the mean values by a very small difference, $E[d_{it}] - \mu_i^* \leqslant \varepsilon \mu_i^*$, i.e., $\varepsilon = 0.001$. The literature shows that the mean (expected) value of the sum of the independently distributed log-normal distributions converges (Beaulieu et al., 1995).

The number of FSA vehicles parked (V_{it}) by industry i during the time period δ is estimated by adding the FSA trips during the current interval t to the vehicles existing from the previous time intervals. The parking duration (D_{it}) is divided into the number of threshold periods the vehicle is parked, given by the ceiling function $\lceil D_{it}/\delta \rceil$ which gives the smallest integer greater than D_{it}/δ. The FSA_{it} is added to the parking demand for the periods the vehicles are parked, i.e., if it is less than the total time interval, $t + \lceil D_{it}/\delta \rceil \leqslant T$. The final outputs of the models are: (i) FSA vehicle parking demand for industry i, for each time interval t, or S_{it}; (ii) parking length PL_t, assuming an average parking length L; and (iii) the Occupancy Rate OR_t defined as the ratio between the PL_t and the existing parking space or the curb space.

5. Numerical results

This section discusses the results obtained from applying the simulation model to hypothetical scenarios. The authors tested the model using a case study from retail and food establishments located in two highly congested blocks in downtown Manhattan. The results were then expanded to the top five ZIP codes in New York City, located in Midtown Manhattan. The analysis was expanded to selected top ZIP codes from six additional American cities. The results were generated for the ZIP code with the highest FSA at each one of these cities.

5.1. Simulation parameters

The authors estimated the simulation parameters as those that best represent the realities of American cities. The simulations use:

- 30-minute time intervals, $\delta = 30$ min.
- Length of FSA standard vehicle parking space, L. For freight pick-ups and deliveries, average length $L = 9.14$ is considered based on the standard length for truck parking (Qcode, 2020). For service vehicles the authors assume $L = 6.7$ m based on the standard commercial and private on-street accessible parking lengths (Minnessota State Department of Transportation, 2015).
- Curb space measured in meters using GIS software for the geographical area studied.
- Daily FSA trip estimates from FTG/STA models from the authors' previous research (Holguín-Veras et al., 2017a; Holguín-Veras et al., 2017b).

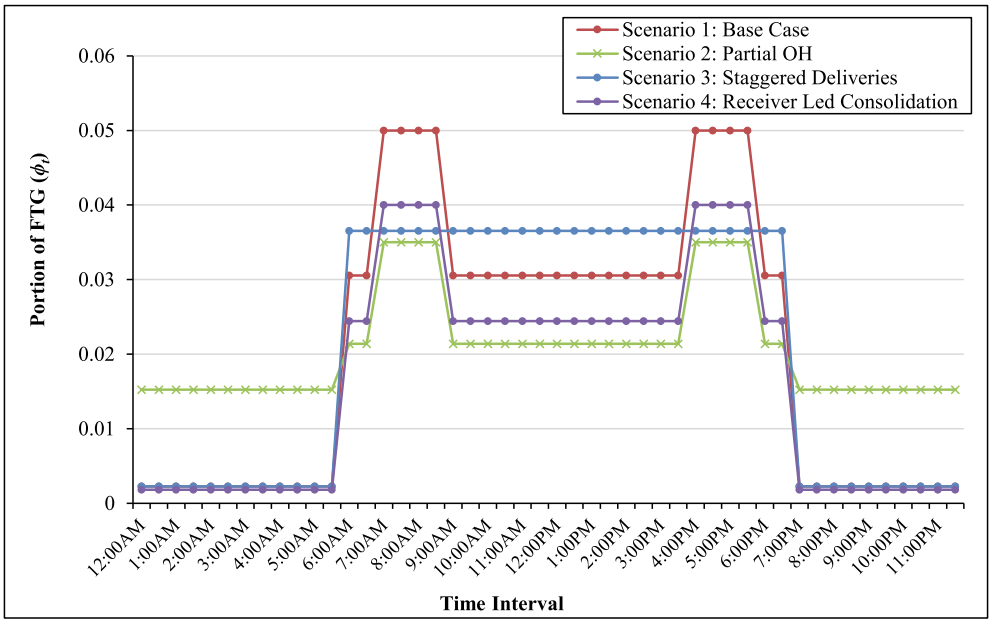


Fig. 5. Traffic Distribution for Freight Pickups and Deliveries in a Typical Day.

- FSA by time-of-day, ϕ_t, follows a distribution as shown in Fig. 5 (freight pick-ups and deliveries) and 6 (service visits), which refers to the proportion of daily FSA trips at each interval.
- FSA durations mean and standard deviation per industry sector from empirical research. Table 3 shows the statistics for service duration using the different sample groups and industry sectors. Table 3 also shows the statistics for the duration of freight pickups and deliveries after subtracting the average 2.5 walking times of the freight parking estimates from (Schmid et al., 2018).
- Walking times to receiver are uniformly distributed,
  - o Freight pick-ups and deliveries, $w_i \tilde{U}[1, 2.5]$
  - o Service visits, $w_i \tilde{U}[1, 2.5]$

The authors selected the parameters of the walking time distribution w_i based on the range of values suggested by freight carriers. The lowest value represents the case where the carrier can park in front of the service activity site to deliver a small package that can be easily retrieved. The upper value represents the maximum time carriers are typically willing to walk to make a single delivery, which is determined by where they are parked. In the case of freight pick-ups and deliveries, a walking distance too far away will prevent carriers from completing the rest of the deliveries. Walking one long block (185 m) results in approximately 2.5 min to the establishment. In the case of service visits, providers may have higher flexibility, thus allowing longer walking times (e.g., 5 min or 375 m). For both cases, the authors assume 4.5 km/h (Muñuzuri et al., 2012), or $w_{UB} = 2.5$ minutes.

FSA trip estimates result from the FTG and STA models from the establishment-level surveys conducted in two US metropolitan areas: New York City (NYC) and the New York State Capital Region (CR) (Holguín-Veras et al., 2017a; Holguín-Veras et al., 2020). The SoHo case discussed in Section 5.2 uses the employment data from 22 establishments to estimate daily FSA trips. The ZIP-code level estimates of cities addressed in Section 5.3 use the County Business Patterns data to generate the FSA estimates at the ZIP-code level using the Freight and Service Trip Generation Software (Holguín-Veras et al., 2017b).

The proportions of daily FSA, ϕ_t account for the time-of-day arrival patterns of commercial vehicles in American cities. Using data from previous studies, the authors estimated the ϕ_t factors for every 30 min of an entire 24-hour-day. Figs. 5 and 6 show the commercial traffic patterns for freight and service, respectively. The authors considered alternative scenarios. The second scenario considers an increase of traffic during the off-hours. The third scenario assumes that pick-ups, deliveries, and service visits are staggered throughout the daytime. The fourth scenario shows a potential reduction of freight trips due to receiver-led consolidation strategies.

Fig. 5 shows the arrival patterns for freight pick-ups and deliveries. The vehicle arrivals consider two peak-hour periods and an off-hour period. The two peak hours (PH) during regular hours include one in the morning (7AM to 9AM) and the other in the evening (4PM–6PM) (Holguín-Veras et al., 2011b; Jaller et al., 2013a). The off-hour period is from 7 PM to 6 AM. Scenario 1 considers 5 % of total traffic during the off hours, 40 % during peak hours, and the rest (47.5 %) spread out throughout the daytime (6-7AM, 9AM-4PM, 6-7PM). Scenario 2 considers 30 % of total traffic during the off hours (Holguín-Veras et al., 2008), 32 % during peak hours, and the rest (38 %) spread out in the daytime. Scenario 3 considers staggered deliveries, i.e., an equal proportion of freight trips during the daytime (6AM–7PM). Scenario 4 considers the receiver-led consolidation strategies, which assume a reduction of 20 % of freight traffic, following the 20–30 % estimates from Holguín-Veras and Sánchez-Díaz (2016).

Fig. 6 shows the service vehicles' arrival patterns. The base-case scenario (Scenario 1) considered the regular commercial vehicle

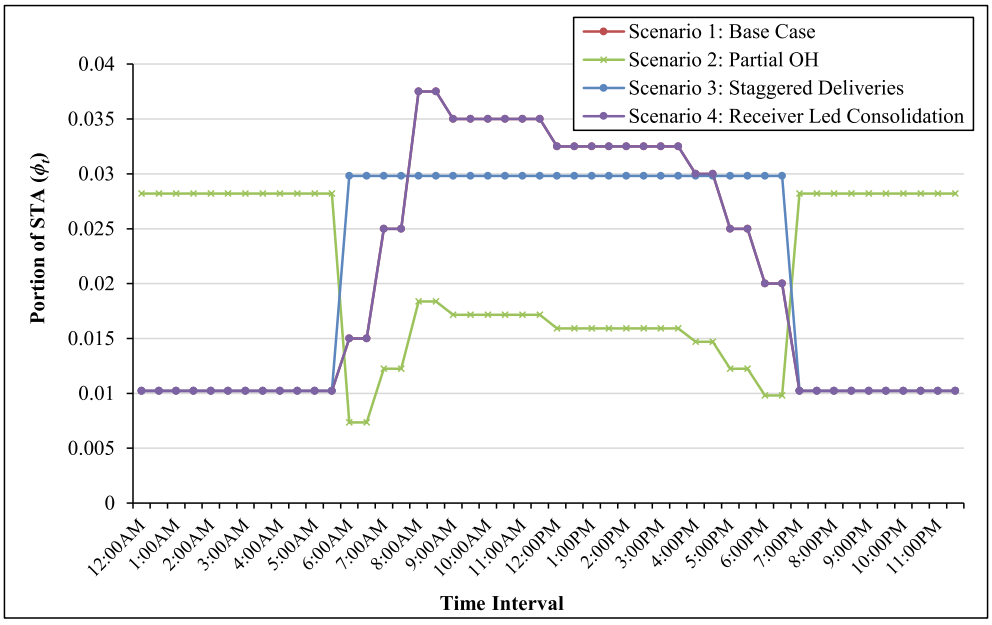


Fig. 6. Traffic Distribution for Service Vehicles in a Typical Day.

time-of-day patterns from previous research (Boile et al., 2004). Scenario 1 assumes 22.5 % traffic during the off hours (7PM–6AM), and the rest (77.5 %) is spread throughout the day, as shown in Fig. 6. Scenario 2 increases to 51 % traffic during the off hours. The 51 % shift corresponds to the authors' empirical analysis about willingness-to-accept off-hour services (Holguín-Veras et al., 2020). Scenario 3 considers an equal proportion of service visits (6AM–7PM) throughout the day. Scenario 4 does not apply to service visits because the latter is not possible in some cases. Thus, scenarios 1 and 4 are the same for service vehicle trips.

To evaluate occupation rates, the simulation model uses different on-street parking capacities. These occupation rates are obtained from road length estimates using GIS software (i.e., TRANSCAD), as explained in Section 4.3. A percent of the road length is considered "available" by visually analyzing typical street lengths. These values still overestimate road lengths, as they do not consider commercial parking regulations and private vehicle parking. Previous research shows that the actual capacities are considerably lower. Chen et al. (2017) explain that street parking turnover rates are extremely low, particularly in congested cities like NYC. As a result, commercial vehicles cannot access legal on-street parking, so they frequently double park to serve nearby customers.

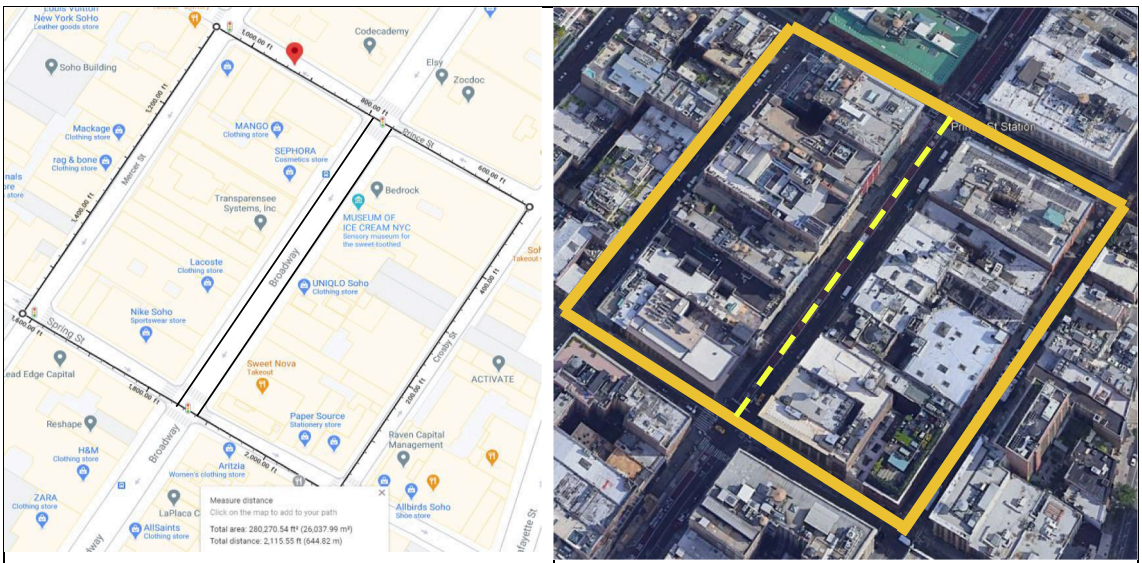
The following case analyzes the FSA parking in a street corridor of a congested area, a busy commercial area in Lower Manhattan, NYC.

5.2. Case study: SoHo BID Manhattan

The case study discussed in this section considers a two-block commercial area in the SoHo business improvement district (BID) in Lower Manhattan, NYC. The two blocks are located along Broadway Avenue, between Prince St. and Spring St. The authors used the data collected from twenty-two commercial establishments to test the parking simulation model. These establishments belong to two industry sectors, the Retail Trade (NAICS 44) and Food and Accommodation (NAICS 72) sectors. The case follows the study from (Campbell et al., 2018). The authors used the latest establishment-level models (Holguín-Veras et al., 2017a) to estimate daily freight and service vehicles resulting in FSA parking needs during peak-hour traffic. Due to the current lack of parking and loading/unloading areas for the establishments studied, all vehicles are forced to search for parking along the curb. Fig. 7 shows the area studied. The figure to the right, Fig. 7(b), indicates the satellite image of the two blocks where the establishments are located. Note that all establishments have their storefront along Broadway Avenue, shown by the dotted line.

The authors estimated the total road length and the length available for parking based on the streets indicated in Fig. 7(a). The total road length of the surrounding streets (Prince St., Mercer St., Spring St., Crosby St.) measured from the start of the block to the end of the intersection is 644.82 m. Note that the parking of vehicles is typically available along these streets on one side of the road, as shown in Fig. 8. Along Broadway Avenue, there are 300 m total road length (150 m per side). Thus, the total road length is 944.82 m. Using Google Maps, the authors estimated that only 210 m of the total road length was usable for on-street parking (i.e., without the intersections, crosswalks, fire hydrants, bus turnover lanes, and other dedicated lanes for turning movements).

To generate the parking estimates for the SoHo case, the authors use the duration statistics for the NYC sample shown in Table 3. Table 4 shows the simulation results for the 22 establishments. These establishments generate nearly 260 daily freight pick-ups and deliveries, accumulating up to 18 vehicles in a single timeslot. The establishments attract close to eight service visits, which may cause the accumulation of two vehicles in a single timeslot during the day. The maximum on-street parking space is 170 m in terms of road occupancy. Recall that the model parameters consider a parking space for freight vehicles of 9.14 m in length and a parking spacing for



(a) SoHo BID Area Studied | (b) Aerial Image of the Area Studied

Fig. 7. Commercial Area in the SoHo BID used in the Case Study.

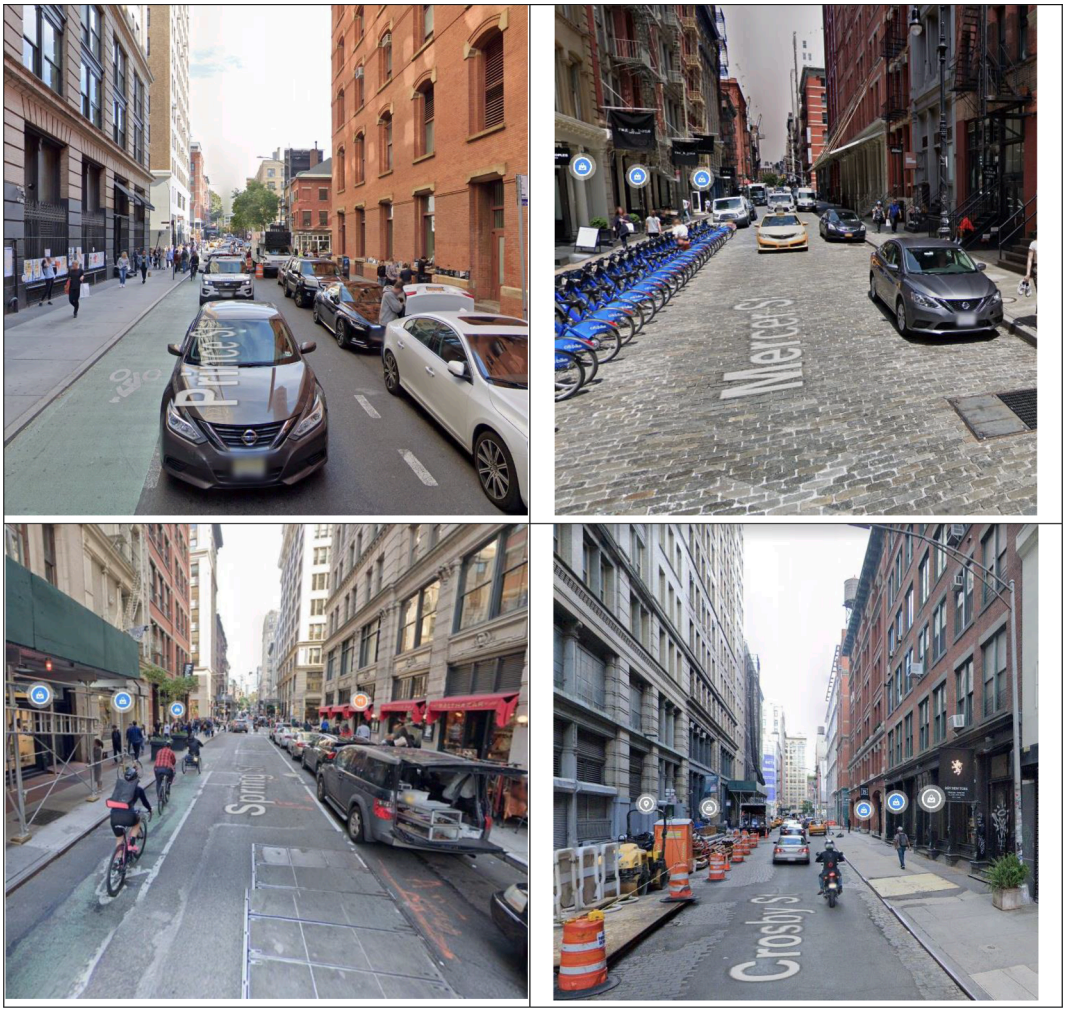


Fig. 8. Street Views of Prince St., Mercer St., Spring St., and Crosby St.

service vehicles that measures 6.7 m in length. The results would show a rate of 82 % occupancy of the actual street capacity if 100 % of the available road length was dedicated for freight and service vehicles. If 75 % of actual street capacity is dedicated to commercial vehicles, the occupancy rate will increase to 109 % during peak hours. These values grow to more than 300 % as dedicated road space for FSA parking decreases to 25 %. These results indicate that under scenario 1, there is a lack of curbside space for freight and service vehicles in the area if current parking capacities are not dedicated entirely to FSA activities. Scenario 2, which increases the deliveries and services in the off-hours, significantly improves occupancy rates. For the case where all road space is dedicated to FSA vehicle

Table 4
Summary Results for SoHo BID.

Indicators	Scenario 1	Scenario 2	Scenario 3	Scenario 4
FTG/day	257.77	257.77	257.77	206.21
STA/day	7.72	7.72	7.72	7.72
Max FTG Vehicles	18	15	16	15
Max STA Vehicles	2	2	2	2
Max Total Vehicles	19	16	17	16
Max FTG Road Length (meters)	164	137	146	137
Max STA Road Length (meters)	13	13	13	13
Max Total Road Length (meters)	171	144	153	144
On-Street Road Capacity	210	210	210	210
Occupancy Rate (100 % road capacity)	0.82	0.68	0.73	0.68
Occupancy Rate (75 % road capacity)	1.09	0.91	0.97	0.91
Occupancy Rate (50 % road capacity)	1.63	1.37	1.46	1.37
Occupancy Rate (25 % road capacity)	3.26	2.74	2.91	2.74

parking, the occupancy rate is 68 % during peak hours. Staggered deliveries (Scenario 3) may not improve as much, with an occupancy rate of 73 %. A freight consolidation among receivers (Scenario 4) yields a similar result to the off-hour deliveries initiative (Scenario 2). The latter indicates that even if time-of-day delivery patterns remain the same, consolidating deliveries (such that the daily trips reduce by 20 %) will reduce peak-hour traffic parking needs as much as the off-hour delivery program.

Table 4 shows the maximum number of parked vehicles during a specific timeslot. Fig. 9 shows this accumulation throughout the day. Fig. 9 shows parking needs (measured in meters) and occupancy per period. Scenario 1 shows two peak-hour periods, one in the morning (with 75 % on-street parking capacity) and one in the afternoon (82 % on-street parking capacity). During the afternoon peak-hour traffic, the number of parked vehicles reached its maximum from 4:30 to 5:00PM. During this period, 19 FSA vehicles required parking. If these were to be lined up in the street, they would occupy 170 m of curb space. Scenario 2 shows an improved situation, where a shift of freight deliveries and services to the off-hours reduces the accumulation of parking during the peak-hour periods. These reductions are also observed now at the 5:30–6:00PM interval, with a maximum occupancy of 68 %. Scenario 3 shows a more uniform distribution, generally derived from the assumption of staggered arrivals. Yet, an interesting finding is that this initiative will not significantly impact the parking situation during daytime hours. The results show that the needs are different from the base case. Scenario 3 experiences their peak close to noon with 73 % parking occupancy and 153 m. Additionally, they experience high occupancy throughout most of the afternoon, with 63 % occupancy. Scenario 4 shows the same maximum parking occupancy of 68 % as Scenario 2 but at 4:30–5:00 pm.

5.3. FSA parking requirements at congested US ZIP codes

This section presents the simulation procedure applied to larger networks, where the authors studied the accumulation of commercial vehicles at the ZIP-code level. The analysis first studied the ZIP codes in New York, NY, with the highest FSA traffic. A second analysis focused on selecting the ZIP codes for other US cities with the highest FSA. The FTG and STA estimates were obtained at the ZIP-code level using the Freight and Service Trip Generation Software (FASTGS) (Holguín-Veras et al., 2017b). The FASTGS uses publicly available employment data from the ZIP Code Business Patterns (CBP) database (U.S. Census Bureau, 2018a). Using 2016 establishment and employment data from the CBP database, the software generated daily FTG and STA estimates by industry sector at the ZIP-code level.

In addition to FTG at the establishment level, the freight pick-up and deliveries included the estimates of trips generated due to internet deliveries to households. The authors compute these estimates using the ratio of internet purchases to households from the National Travel Household Survey (McGuckin and Fucci, 2018). The authors calculated B2C vehicle trips by assuming an average of five deliveries per vehicle trip and added them to the FTG totals.

Using GIS software, the authors estimated total road network lengths at these ZIP codes as explained in Section 4.3. The authors estimated how much of the road could be used for parking at the curb. After identifying typical commercial streets at each of these ZIP codes, the study used Google Maps to obtain an approximate value of available parking length. The available parking length is estimated by subtracting the total road length from the space for the intersections, crosswalks, fire hydrants, bus turnover lanes, and other dedicated lanes for turning movements. According to US regulations, the fire hydrants occupy at least 15ft to each side.

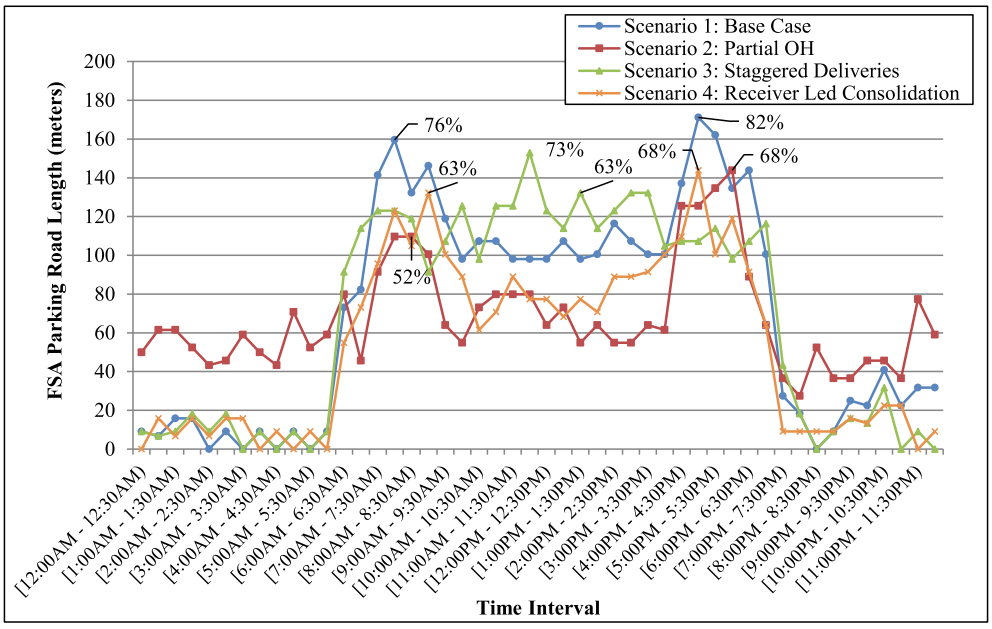


Fig. 9. Traffic Distribution for Service Vehicles in a Typical Day.

5.3.1. Top ZIP codes in New York City

The analysis at the ZIP-code level starts with the top ZIP codes in New York, NY. The authors selected five ZIP codes with the highest FSA in New York, in Midtown Manhattan (i.e., 10001, 10018, 10019, 10022, 10036). ZIP code 10001 includes the buildings that have their own ZIP code (i.e., 10118, 10119, 10120, 10121, 10122) but that are within the boundary of 10001.Similarly, ZIP code 10019 includes several buildings among their blocks (i.e., 10020, 10103, 10104, 10105, 10106, 10107, 10111, 10112). ZIP code 10022 covers other buildings with their own ZIP code that ranges from 10151 to 10155. The area covered by ZIP code 10036 also includes a building with ZIP code 10110. Fig. 10 shows the ZIP codes included in the analysis. These were the ZIP codes with the highest daily commercial vehicle traffic.

Table 5 shows the FTG and STA model results from the NYC models implemented in the FASTGS software (Holguín-Veras et al., 2017a; Holguín-Veras et al., 2020). Approximately 33, 300 establishments and 816, 250 employees in the area covered by the five ZIP codes generate about 146, 700 daily vehicle trips. The latter includes 131, 200 freight pick-ups and deliveries, 12, 235 service vehicles, and 4, 230 e-commerce trips (B2C) to households. The latter represents the bulk of daily traffic that will compete for parking space in Midtown Manhattan.

The top FSA generator is ZIP code 10001, where the Empire State Building and other iconic buildings are located. The largest Macy's store, which covers an entire block, is also in this ZIP code. Approximately 8, 200 establishments and 175, 500 employees at this ZIP code generate close to 36, 000 FSA daily vehicle trips. About 760 vehicle trips are added due to e-commerce purchases. The west side of Midtown Manhattan belongs to ZIP codes 10018, 10036, and 10019. A large neighborhood known as Hell's Kitchen or Clinton is located within these ZIP codes. This area provides transportation, medical, and warehouse support to Manhattan's business district. ZIP code 10018 houses 6, 580 establishments that generate approximately 33, 000 FSA trips and 300 e-commerce trips to households. ZIP code 10036 has a larger residential neighborhood than 10018, where households generate about 850 e-commerce trips. In addition, establishments generate 29, 000 freight pick-ups and deliveries and 2, 450 service trips daily. ZIP code 10019 has around 5, 600 establishments that generate close to 22, 400 FSA trips daily. ZIP code 10022, located on the eastern side of Midtown Manhattan, generates approximately 23, 500 FSA trips, including 980 e-commerce trips daily.

Table 6 shows the base case results and three policy scenarios (off-hour FSA, staggered freight deliveries, and receiver-led consolidation). The authors analyze three instances for the percentage of curb space allotted to FSA (100 %, 75 %, and 50 %). The results indicate the maximum FSA parking and parking occupation accumulation per ZIP code. Under base case conditions, there are approximately 2, 400 parking slot requirements at ZIP code 10001, which spans 21 km in road length. ZIP codes 10018 and 10036 each accumulate about 2, 100 parking spaces, spanning 19 km in road length. ZIP codes 10019 and 10022 also generate a significant daily accumulation of parked vehicles during peak hours, requiring between 12 and 13 km of road length at each ZIP code.

The table also indicates the maximum capacities of the network during peak time. The most critical are the first three ZIP codes, i.e., 10001, 10018, and 10036, which occupy more than 100 % of the road network under base case conditions. If all curb space was

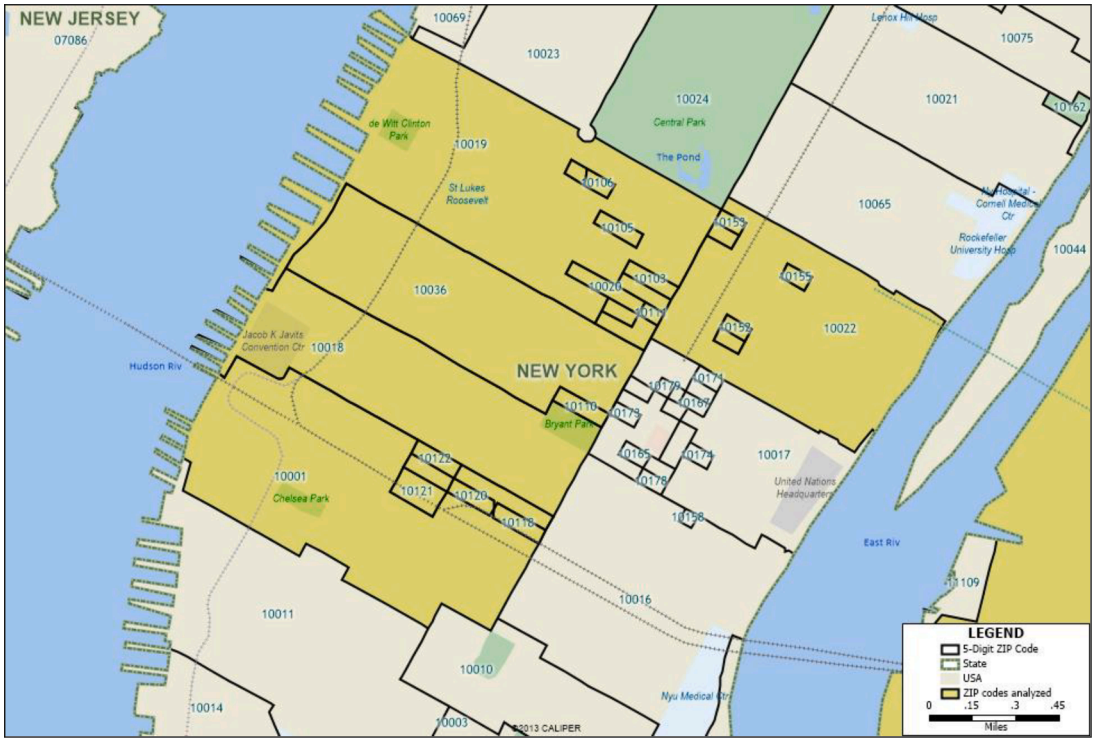


Fig. 10. NYC's ZIP Codes with Highest FSA.

Table 5
FTG and STA Estimates at Largest ZIP Codes in NYC.

ZIP Code(s) analyzed	10001*	10018	10036*	10022*	10019*
Population (2016)	23, 947	9719	26, 868	30, 807	41, 592
Total linear road cap (km)	21.16	18.77	16.88	21.27	26.55
Establishments	8, 198	6, 580	6, 208	6, 672	5, 633
Employment	175, 518	143, 757	164, 025	154, 584	178, 357
FTG/day	32, 684	31, 079	29, 184	19, 931	18, 320
STA/day	2, 501	1, 929	2, 451	2, 628	2, 726
Internet Deliveries to HH	3, 808	1, 545	4, 272	4, 898	6, 613
E-commerce trips	762	309	854	980	1, 323
FTG + B2C	33, 446	31, 388	30, 038	20, 911	19, 643
FSA/day	35, 947	33, 317	32, 489	23, 539	22, 370

*ZIP code includes large buildings that have their own ZIP code: ZIP code 10001 includes 10118, 10119, 10120, 10121, 10122. ZIP code 10036 includes 10110. ZIP code 10022 includes 10151–10155. ZIP code 10019 includes 10151–10155. ZIP code 10019 includes 10020, 10103, 10104, 10105, 10106, 10107, 10111, 10112.

Table 6
Peak Hour FSA Parking Needs at Top NYC ZIP Codes.

Scenario 1: Base Case

Zip Code	FTG	STA	Parked Vehicles	Road Length(km)	Occupancy-100 %	Occupancy-75 %	Occupancy-50 %
1001	1544	32	2374	21.13	99.82 %	133.10 %	199.65 %
10018	1485	33	2165	19.37	103.19 %	137.59 %	206.39 %
10036	1423	52	2162	19.20	113.75 %	151.67 %	227.51 %
10022	918	76	1592	13.84	65.05 %	86.73 %	130.10 %
1019	878	74	1546	13.42	50.52 %	67.36 %	101.04 %

Scenario 2: Partial Off Hour Deliveries and Services

Zip Code	FTG	STA	Parked Vehicles	Road Length(km)	Occupancy-100 %	Occupancy-75 %	Occupancy-50 %
10001	1081	24	1636	14.60	68.98 %	91.97 %	137.96 %
10018	1040	25	1514	13.63	72.61 %	96.81 %	145.22 %
10036	996	39	1504	13.41	79.43 %	105.91 %	158.86 %
10022	642	57	1082	9.54	44.87 %	59.82 %	89.73 %
10019	615	56	1042	9.14	34.42 %	45.90 %	68.85 %

Scenario 3: Staggered Deliveries

Zip Code	FTG	STA	Parked Vehicles	Road Length(km)	Occupancy-100 %	Occupancy-75 %	Occupancy-50 %
10001	1128	26	1849	16.30	77.00 %	102.67 %	154.00 %
10018	1085	26	1671	14.85	79.10 %	105.47 %	158.20 %
10036	1040	41	1685	14.83	87.84 %	117.12 %	175.68 %
10022	671	61	1278	11.01	51.74 %	68.99 %	103.48 %
10019	642	59	1213	10.40	39.16 %	52.21 %	78.32 %

Scenario 4: Receiver Led Consolidation of Freight Deliveries

Zip Code	FTG	STA	Parked Vehicles	Road Length(km)	Occupancy-100 %	Occupancy-75 %	Occupancy-50 %
10001	1235	32	1973	17.48	82.58 %	110.10 %	165.15 %
10018	1188	33	1788	15.91	84.78 %	113.04 %	169.55 %
10036	1138	52	1804	15.89	94.11 %	125.48 %	188.22 %
10022	734	76	1337	11.54	54.27 %	72.37 %	108.55 %
10019	702	74	1293	11.10	41.80 %	55.74 %	83.61 %

dedicated to FSA vehicles, ZIP codes 10018 and 10036 would be over capacity with 103 % and 113 %, respectively. If 50 % of the curb space was dedicated to FSA vehicles, none of these ZIP codes would have enough capacity. The occupancy rates at these ZIP codes align with the findings from the previous studies, which ranged between 90 % and greater than 100 % depending on the time of the day (Schaller et al., 2010; Jaller et al., 2013a).

Table 6 shows that all three initiatives effectively reduce occupancy to less than 100 % if all of the streets were available for FSA vehicle parking. Implementing the partial off-hour delivery initiative (Scenario 2) is most effective, as it considerably reduces FSA parking needs during peak hours at all ZIP codes (reductions range from 30 to 32 %). Occupancy at ZIP codes 10001 and 10018 drops to approximately 70 %, while ZIP Code 10036 drops to 80 %. The latter is if all available on-street parking was available for FSA vehicles. If 75 % of the road were made available, occupancy would increase to 92 % at ZIP code 10001 and 97 % at ZIP code 10018.

ZIP code 10036 will be over capacity (106 %) even if partial deliveries and services were shifted to the off-hours (Scenario 2). The last two ZIP codes would have less than 60 % occupancy if 75 % of all available road was dedicated to FSA vehicle parking. The staggered deliveries and service visits (Scenario 3) are less effective (20–22 % reduction in parking spaces) and cannot accommodate all FSA vehicles during peak hours. The receiver-led consolidation initiative is less effective for three out of five ZIP codes, particularly 10018 (only an 18 % reduction). Yet, for ZIP codes 10019 and 10022, this initiative (Scenario 4) is better with reductions of 24 %, a little less than the staggered deliveries and services initiative (Scenario 3).

The results at the top ZIP codes in Midtown Manhattan illustrate the magnitude of the parking issue when there is not enough curb space in dense neighborhoods. Especially critical is the area where the Lincoln Tunnel bridge traffic enters (ZIP code 10001), and most of the west side neighborhood in Manhattan (ZIP codes 10018 and 10036). Compared to ZIP code 10001, the area covered by ZIP codes 10018 and 10036 has less curb space for on-street parking and still requires approximately 2, 100 vehicles during peak hours. In normal conditions, parking occupancy during AM and PM peak hours was over 100 % if all curb space was dedicated to FSA activities. These values increase more than 200 % when dedicated space drops by 50 %. The only initiative that reduces occupancy to less than 100 % is the partial off-hour deliveries and services.

Fig. 11 shows the time-of-day parking needs at ZIP code 10018 to illustrate the parking patterns at these four scenarios. The results indicate the effectiveness of each demand management strategy compared to the base case. The initiative that works best is the off-hour deliveries (Scenario 2), which reduced the parking by 29.6 %, while staggered deliveries (Scenario 3) reduced occupancy by 23.3 %. The least effective in this case was receiver-led consolidation (Scenario 4), which reduced occupancy by only 17.8 % when compared to the base case.

5.3.2. *Top ZIP codes in other US cities*

The simulation model studied the accumulation of commercial vehicles at the most congested ZIP codes in large American cities. The selection of cities was based on the population density ranking of cities with populations larger than 100, 000 (U.S. Census Bureau, 2018b; World Population Review, 2019). The NYC model was used for cities with a population larger than 750, 000 (i.e., Austin, Columbus, and San Jose). The combined data (NYC + CR) model was used for cities with a population between 250, 000 and 750, 000 (Seattle, Washington, and Boston). The CR model was used when the cities' populations were less than 250, 000. The freight demand models' transferability was tested by Holguín-Veras et al. (2013).

Table 7 shows the daily FSA traffic estimates for the selected ZIP codes. The ZIP codes chosen for each city produced the highest freight and service trips. These ZIP codes were typically located close to the downtown areas.

The cities in Table 7 are organized in ascending order concerning the density of the entire city. The ZIP code 78701 in Austin, TX, surrounds the Texas Capitol and is where the Medical, Red River Cultural, and Rainey Street Historical districts reside. The ZIP code also includes a couple of busy commercial districts, mainly the Warehouse and Second Street districts. In Austin's ZIP code, nearly 3000 establishments and 60, 000 employees generate 8, 250 daily FSA vehicle trips. Additionally, the residential buildings attract 158 daily e-commerce trips. Columbus, OH's busiest ZIP code (43215) is where the Convention Center and Arena are located. The Columbus State Community College and busy marketplace are also part of the ZIP code. At this ZIP code, 2, 571 establishments generate 7, 200 daily FSA vehicle trips. These estimates increase by 287 when e-commerce vehicle trips to households are added.

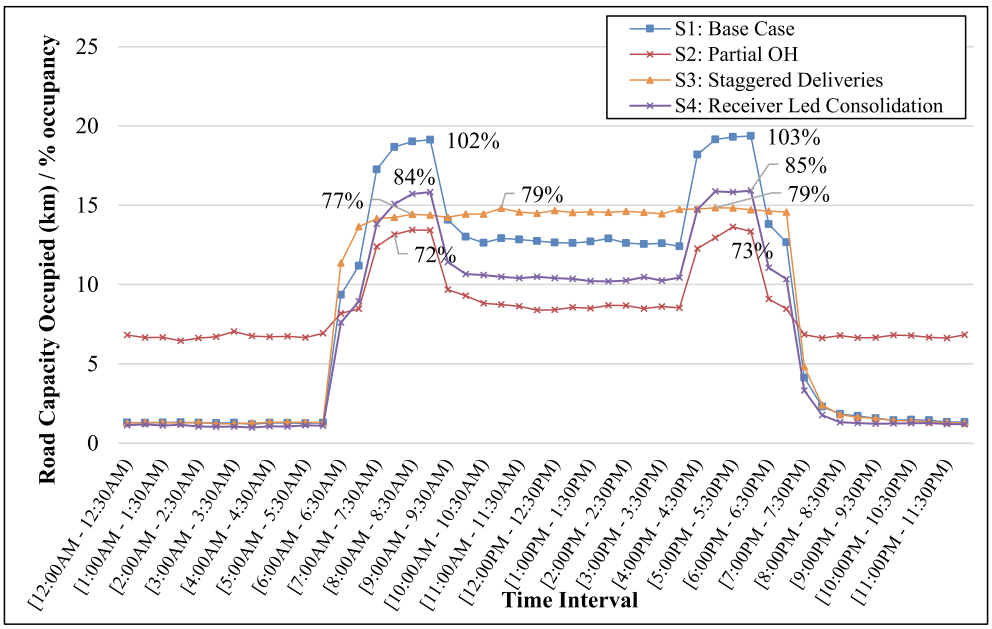


Fig. 11. Time-of-day Parking Capacity and Occupancy for ZIP Code 10018.

Table 7
FTG and STA Estimates at the Largest ZIP Codes in Other US Cities.

	Austin, TX	Colum-bus, OH	San Jose, CA	Seattle, WA	Washing-ton, DC	Boston, MA
Population (2016)	916, 906	852, 144	1,023, 031	688, 245	672, 391	669, 158
Population density (pop/km2)	1103.6	1505.1	2225.1	3169.8	4246.2	5344.3
ZIP Code(s) analyzed	78701	43215	95112	98101	20036	02116*
Population at ZIP Code(s)	7875	14, 322	62, 039	12, 408	5836	23, 215
Total linear road capacity (km)	42.2	159.7	143.5	25.7	14.5	37.4
Establishments (All Sectors)	2, 996	2, 571	2, 197	2, 956	2, 824	2, 060
Employment (All Sectors)	59, 387	67, 682	44, 735	87, 499	65, 962	61, 340
FTG (All Sectors)/day	7, 279	6, 274	11, 071	9, 537	4, 458	7, 372
STA (All Sectors)/day	970	925	747	1, 583	1, 474	1, 180
Internet Deliveries to HH/day	788	1, 432	7, 569	1, 241	584	3, 157
B2C/day (5 deliv. to HH/trip)	157.50	286.44	1, 513.75	248.16	116.72	631.45
FTG + B2C/day	7, 437	6, 560	12, 585	9, 785	4, 575	8, 003

*In Boston, this ZIP code includes a large building that have its own ZIP code, 02117.

The third ZIP code studied is 95112 in San Jose, CA. This ZIP code includes several neighborhoods, like Japantown. It also houses part of San Jose State University, and busy commercial districts. Approximately 2, 200 establishments generate 11, 000 pick-ups and deliveries, and approximately 750 service visits.

Seattle, WA's ZIP code 98101 surrounds the Pike Place Marketplace and numerous retail establishments. The ZIP code houses close to 3, 000 establishments that generate 9, 500 vehicle trips and 1, 500 service visits daily. The residential neighborhood also attracts more than 100 e-commerce trips. ZIP code 20036 in Washington DC is around Dupont Circle, where several embassies are located. The area is surrounded by restaurants and hotels and is where the National Geographic Museum is located. As a more service-oriented city, the ZIP code is home to 2, 800 commercial establishments, from which 2, 400 belong to the service sector. All 2, 800 establishments generate 5, 930 daily vehicle trips.

The largest ZIP code in Boston, MA (02116) also includes a large building with its own ZIP code (02117). Like in Manhattan, these ZIP codes are usually large generators of freight and service activity. In Boston, MA, ZIP code 02116 includes the Commonwealth Avenue Mall, the Boston Public Library, and several restaurants at Boston's Park Plaza. At these ZIP codes, 2, 060 establishments generate 8, 550 daily FSA vehicle trips. Internet deliveries to households add approximately 630 daily vehicle trips.

The duration parameters used for the simulation are those shown in Table 3. Similar to the application of the FTG and STA models, the duration parameters used for the analysis of different cities corresponded to the city's size. Based on the NYC sample, the large city estimates were used for cities with populations larger than 750, 000 (i.e., Austin, Columbus, and San Jose). The combined data (total sample) was used for cities with a population between 250, 000 and 750, 000 (Seattle, Washington, and Boston).

Fig. 12 shows the results in terms of road length requirements (in kilometers) for the different ZIP codes by time-of-day in the base case scenario. The patterns show maximum parking needs during the morning (7:30 to 9:00AM) and afternoon (4:00 to 6:00PM) peak periods. The largest peak is observed in San Jose's ZIP code, which can require up to 8 km of curb space to accommodate FSA parking. The needs can be substantially reduced through the implementation of demand management strategies.

Table 8 presents the results for the base case (scenario 1) and the off-hours initiative (scenario 2). The results are in descending order from the ZIP code with the highest on-street parking requirements (Seattle, WA) to the less critical (Columbus, OH). The table shows the vehicle trips and maximum parked vehicles for freight pick-ups and deliveries (FTG column) and service visits received (STA column). The results also show the total parked vehicles and road length in meters required at the ZIP-code level in each city. Except for Seattle and Washington DC, the cities can accommodate FSA parking under base conditions, even if only 25 % was dedicated for FSA vehicles. The most congested ZIP codes in Seattle and Washington DC can still accommodate vehicles on-street if enough space is dedicated to FSA vehicles. At 25 %, the ZIP codes in Washington D.C. and Seattle have 110 % and 109 % of occupancy, respectively. Thus, under the current conditions, the results suggest that a little more than 25 percent should be dedicated to FSA on-street parking for these urban spaces.

Table 8 also shows Scenario 2, which considers an increase in the proportion of traffic during the off-hours. Compared to the other two scenarios, this was most effective, as indicated by the NYC ZIP codes analysis in the previous section. The off-hour deliveries initiative improves the outcome of parking availability, even if there is not enough space dedicated to FSA vehicles. In general, if 25 % of the curb space was dedicated to FSA parking, the freight and service vehicles would find parking spaces at the curb.

6. Policy implications

The methodology and the simulation procedure proposed in this study help design various urban parking policies aiming to achieve effective freight and service operations. The main advantage of this paper is that these models could quantify the parking requirements separately for freight and services across various industry sectors and times of the day. The authors present a few ideas of such policies below.

Locating/Regulating Dedicated Parking Space for FSA: One key consideration is identifying dedicated space for FSA parking needs. The models help determine areas and times where the road network should reserve on-street space to reduce the overall parking

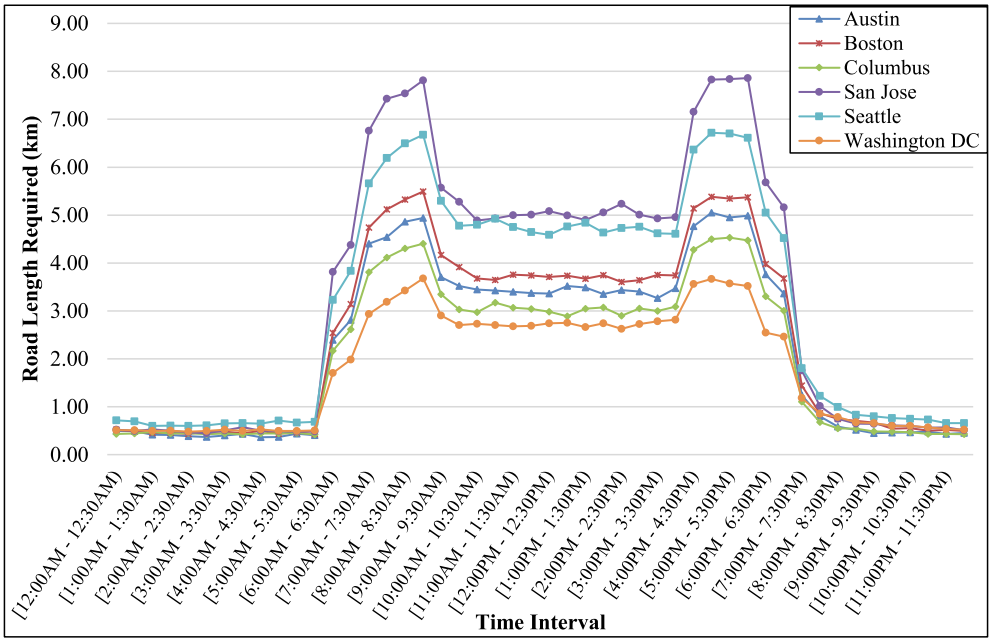


Fig. 12. Time-of-Day Parking Road Length Requirements for City ZIP Codes - Scenario 1.

Table 8
Peak Hour FSA Parking Needs at City ZIP Codes - Scenario 1Scenario 1 (Base Case).

Scenario 1 (Base Case)

City	ZIP Code	FTG	STA	Parked Vehicles	Road Length(m)	Occupancy- 100 %	Occupancy-50 %	Occupancy-25 %
Seattle	98101	436	44	778	6720.52	26.14 %	52.28 %	104.57 %
Washington DC	20036	168	49	444	3678.76	25.32 %	50.64 %	101.29 %
Boston	02116	364	30	632	5491	14.67 %	29.33 %	58.66 %
Austin	78701	312	25	579	5051	11.96 %	23.92 %	47.85 %
San Jose	95112	603	18	878	7857.18	5.48 %	10.95 %	21.90 %
Columbus	43215	279	27	521	4530.14	2.84 %	5.67 %	11.35 %

Scenario 2 (Partial Off-Hour Deliveries and Services)

City	ZIP Code	FTG	STA	Parked Vehicles	Road Length(m)	Occupancy- 100 %	Occupancy-50 %	Occupancy-25 %
Seattle	98101	305	33	524	4544.2	17.68 %	35.35 %	70.70 %
Washington DC	20036	117	37	302	2550.44	17.56 %	35.11 %	70.22 %
Boston	02116	254	23	428	3760.64	10.04 %	20.09 %	40.18 %
Austin	78701	218	19	395	3468.78	8.21 %	16.43 %	32.86 %
San Jose	95112	422	13	588	5288.92	3.69 %	7.37 %	14.74 %
Columbus	43215	195	20	350	3059.92	1.92 %	3.83 %	7.67 %

time for freight and service vehicles. The analysis at the ZIP-code level suggested that highly congested streets like those in the downtown areas should have dedicated spaces for parking freight and service vehicles, at least during the peak-hour traffic. Except for NYC, the ZIP codes in other cities would accommodate FSA needs if at least 25 percent of the on-street road is dedicated to freight and service vehicles. If no regulation exists to accommodate FSA needs, the parking space may be compromised by passenger vehicles that use up significant portions of these spaces, particularly during daytime hours. As observed through Google Maps, most on-street parking is occupied by passenger vehicles at these congested ZIP codes. In addition to locating parking spaces, these models also assist in obtaining the time limits required for parking at various times of the day. For instance, if restaurants dominate an area or ZIP code, the model would estimate the demand and durations concerning the industry sector corresponding to restaurants. In essence, the maximum parking duration limit at various ZIP codes could be estimated from these models. A ZIP code with restaurants would probably require less parking time than a ZIP code with wholesale goods.

Demand Management Strategies: The analysis at highly congested city streets like those in NYC suggest that even with 100 percent dedicated space for FSA, there would still be a lack of on-street parking. Another way of improving the parking of FSA traffic is the implementation of demand management initiatives such as off-hour deliveries, staggered deliveries, and receiver-led consolidation, discussed in this paper. This study shows that all three initiatives successfully reduced FSA traffic during peak hours. Moreover,

the analysis shows that the partial off-hour deliveries and services were the most effective. Shifting 30 % of the freight vehicles and 51 % of the service vehicles to the off-hours improves capacity by 30 %. In NYC, the occupancy reduces from 134 to 94 percent during peak hours at the most critical ZIP code (i.e., 10018). Note that the percentages that were used in this study were not arbitrarily chosen. Previous empirical research developed for freight carriers and receivers indicated that nearly 30 % of the freight pick-up and deliveries could be shifted to the off hours (Holguín-Veras et al., 2017c). The authors' previous research of STA revealed that companies could switch the reception of their services by 51 percent to the off hours (Holguín-Veras et al., 2020).

Developing Parking Infrastructure: In designing freight or service loading or unloading bays, this research would assist in obtaining the optimal number of parking spots that should be constructed or allotted. Promising areas of future research include the revision of construction codes of new commercial complexes, infrastructure development, or land-use planning that incorporate FSA parking needs.

Smart Parking Practices: The advancement in cellphone technologies could enable public authorities to use variable parking signs, and parking time limits depending on the type of activity the commercial vehicle is performing, the industry sector, and the time of arrival. These models help in coming up with reasonable estimates of these practices.

Parking Permits to Occasional Service Vehicles: This research shows service visits, because of their longer durations, require a considerable amount of parking space compared to freight pick-ups and deliveries. A way of reducing service vehicle congestion at the curb is to provide special parking permits for service vehicles. Some cities, like Washington, DC, provide temporary parking permits for contractors and service companies (District Department of Transportation's-DDOT, 2022). Providing temporary parking permits for special service vehicles in areas where parking is prohibited for other vehicles would alleviate the need for space for these more prolonged activities.

Traffic engineers and city officials must consider the impacts of daily freight and service vehicle traffic in the decision-making and implementation of policies. The chief objectives of the paper were to quantify the FSA parking demand and demonstrate the importance of using freight demand management initiatives to reduce peak FSA traffic. Our model insights serve as a tool for any freight or service planning process by quantifying daily FSA parking needs. Nonetheless, these analyses should not overlook private vehicle parking needs, which compete with commercial vehicles for space. Future research should focus on integrating the demand models for passenger and commercial vehicles to better manage the parking needs in urban centers.

Finally, a combination of regulatory and demand management initiatives could improve the use of the curb space in urban areas, thereby reducing traffic congestion and improving the quality of life.

7. Concluding remarks

Allocation of on-street (curbside) space for commercial parking is one of the key challenges faced by urban transportation planners. The previous studies found that freight vehicles often double park or incur parking fines (Jaller et al., 2013a). Although the commercial traffic includes vehicles of various sizes from freight, services, and e-commerce, the modeling techniques developed so far in the literature primarily focus on estimating freight demand at the establishment level or simulating freight parking demand using partially available data. There is hardly any research that considers service trips in the US except for a few recent publications (Holguín-Veras et al., 2020). In addition, estimation of demand (e.g., trips per day) alone is not sufficient for informing parking policies as the duration for which each vehicle parks would affect the number of parking spots required at a given time. The parking durations vary heavily between freight and services, as the latter park longer, across various industry sectors. Hence, it is necessary to combine demand modeling and parking durations for freight, services, and e-commerce to obtain a comprehensive idea of commercial traffic demand. Also, there is a dire need to develop methodologies that policymakers could use to estimate commercial parking demand at the required geographical level (e.g., ZIP code, county, city) without requiring additional resources or data collection efforts. The research presented in this paper is a step towards addressing the issues mentioned above by developing a modeling framework to estimate the urban commercial (freight, services, and e-commerce) parking demand using the publicly available data (U.S. Census Bureau, 2018a).

The commercial vehicle parking simulation model proposed in this paper utilizes the existing demand models (Holguín-Veras et al., 2017a; Holguín-Veras et al., 2020) for freight and service activity (FSA) vehicle trips, along with probability distributions for respective parking durations. The authors tested the model on a small network, two blocks at the SoHo commercial complex in New York City (NYC), as an extension of previous research (Campbell et al., 2018). Upon validation, the FSA parking demand is analyzed for the busiest ZIP codes in New York City as well as in the top ZIP codes of cities across the US (Austin, TX; Columbus, OH; San Jose, CA; Seattle, WA; Washington DC; Boston, MA). The parking simulation is executed for vehicle arrivals every 30 min for five days, and it evaluates three demand management initiatives in addition to the base case. The model runs four scenarios: (1) base case (current); (2) off-hour deliveries and services, which considered a higher share of FSA during off-hours (7 PM- 6 AM); (3) staggered deliveries; and (4) receiver led consolidation strategies. The shares of FSA in the second scenario are obtained based on previous research on willingness-to-accept freight and services during off-hours (Holguín-Veras et al., 2018c; Holguín-Veras et al., 2020). For the fourth scenario, this study assumes a 20 % reduction of freight vehicle trips, based on previous research that estimates between 20 and 30 % of freight traffic potential reductions with the receiver-led consolidation program (Holguín-Veras and Sánchez-Díaz, 2016). Although all three demand management initiatives successfully reduced freight parking needs, shifting FSA activities to the off-hours was the most effective. The results show that moving 30 percent of freight pick-ups and deliveries and 51 % of service visits to the off-hours reduces the parking needs during the daytime by 30 percent. The latter allows more space for FSA parking when passenger vehicles utilize most on-street parking.

Notwithstanding its numerous contributions, the research in the paper has limitations. One limitation is the assumption of walking times from the parking space to and from the service activity site. Due to the lack of data from service providers, the authors assumed a

uniform walking duration with data from past research among freight carriers. A second limitation is related to geography. The freight durations are available only for a big city (NYC), whereas the service activity durations could include smaller urban areas (capital region, CR). These assumptions can be improved by collecting actual walking times from freight and service providers through observational studies or surveys. The third limitation is on the FSA durations by industry sector, as the latter play a huge role in parking durations. For example, a construction vehicle may need to park longer compared to a retail delivery vehicle. The FSA parking durations were incorporated in the simulation by an aggregated level of industry sectors (FIS and SIS) due to insufficient data to obtain the probability distribution at a disaggregated level (e.g., 2-digit NAICS). The fourth limitation is related to parking supply data. Due to a lack of data, actual parking along the curb is not considered. Instead, an approximate road length for parking is estimated from GIS layers of streets in respective ZIP codes. The final limitation is about advanced models for parking durations, i.e., parametric models. Parametric duration models are not included for two reasons. One, the data available is very limited. Second, these models cannot be applied to publicly available data, which is one of the primary motivations of this research.

Despite the limitations mentioned above, this research is a huge leap towards improving the urban commercial freight research to assist urban planners in quantifying parking demand and in designing policies for sustainable commercial operations in US cities.

CRediT authorship contribution statement

Diana G. Ramirez-Rios: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Lokesh Kumar Kalahasthi:** Conceptualization, Methodology, Data curation, Formal analysis, Software, Validation, Writing – original draft, Writing – review & editing. **José Holguín-Veras:** Conceptualization, Project administration, Funding acquisition, Supervision, Resources, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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